

Bilingual Children Speech Classification: Assignment 1

Information Retrieval (5LN712)
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1 Introduction

For this assignment I chose to train three classifiers to predict the L1 of a given text. The text classifiers are trained with CHILDES/CHAT bilingual child speech transcripts. I was motivated to do this topic by recent softwares, like BoldVoice¹, which detect and analyze accents. Furthermore, I am largely interested in speech processing, bilingualism, and how humans and ML systems detect accents, speech impediments, and other linguistic features from spoken language. While this assignment uses transcripts rather than speech, I believe it was a good introduction to using agents and now I can look into expanding the research to speech :D

2 Dataset

I created my own Huggingface Dataset² using an existing CHILDES Kaggle dataset³ as well as synthetically created data using ChatGPT to expand and balance the existing data. I did this since the existing data was predominantly composed of Spanish and Mandarin L1 speakers with little Korean L1 data. So the initial classifiers, trained only on *real* data, had even lower macro f1 scores.

Due to the time constraints of the assignment I used transcripts rather than actual speech files, but in the future I think it would be interesting to train classifiers using only speech data and analyze/classify at the phoneme and word levels. I believe this would lead to better results.

2.1 Dataset Creation

I created a script, with CodeX, to parse the original CHILDES files using the `pylangacq` library. Only child utterances (CHI) were extracted from the transcripts. Also in the pre-processing pipeline, I removed CHAT annotation symbols, empty utterances, and extremely short text segments.

To solve the data imbalance, I generated synthetic data using ChatGPT to create additional utterances that approximately matched the style and fluency patterns of bilingual child speech.

¹<https://start.boldvoice.com/accent-oracle>

²https://huggingface.co/datasets/nimuezorro/bilingual_children_speech

³<https://www.kaggle.com/datasets/rtatman/corpus-of-bilingual-childrens-speech>

3 Model

3.1 Embedding Model

For my embedding model I chose sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 as it produces strong sentence-level semantic embeddings, is trained on many languages, and works well for small embedding-classification projects.

3.2 Classifiers

For my classifiers I chose to use Logistic Regression, Linear SVM, and Random Forest. While these were the classifiers that CodeX suggested, I also found these to be rather common and efficient approaches for embedding-based NLP classification tasks.

4 Results

The Logistic Regression classifier largely outperformed the Random Forest classifier and slightly outperformed Linear SVM. This suggest that simpler linear models may be better suited for this type of embedding-based linguistic classification task than more complex nonlinear approaches. In the table below are accuracy and Macro F1 for each classifier.

Model	Accuracy	Macro F1
Logistic Regression	0.426	0.181
Linear SVM	0.407	0.154
Random Forest	0.269	0.105

Table 1: Classification results on the bilingual speech dataset.

While the classifying task I chose was relatively difficult I believe my small dataset is also to blame for weak classifier performance.

5 Reflection

Overall this was a good introduction to using agents to build fully functional pipelines. I think if you use the agent as an extension of your own knowledge, i.e. telling it what you want and how it should be built, you'll get good results. However, if you aren't sure how your pipeline should function it'll result in wasting tokens and code you don't understand. I found using ChatGPT to first brainstorm the intended goal of the project and overall pipeline was very helpful. I look forward to building my own agent and understanding the mechanics behind the pipeline.